Increased metabolic efficiency via product-dependent enzyme clustering

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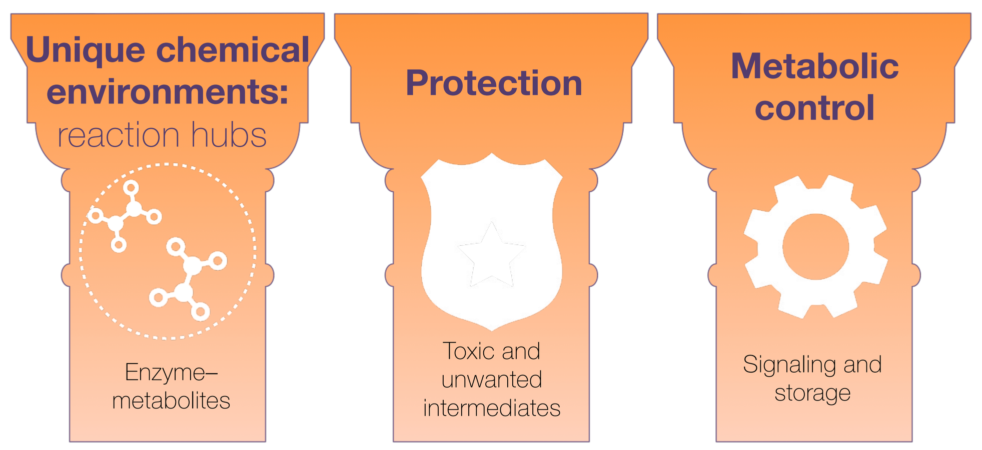
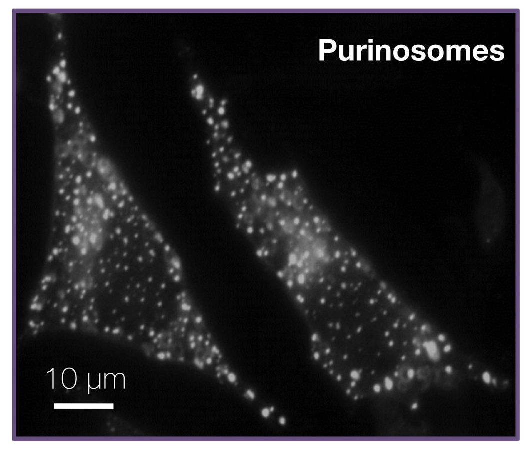
Biomolecular condensates: Metabolic compartmentalization

Three pillars of compartmentalization

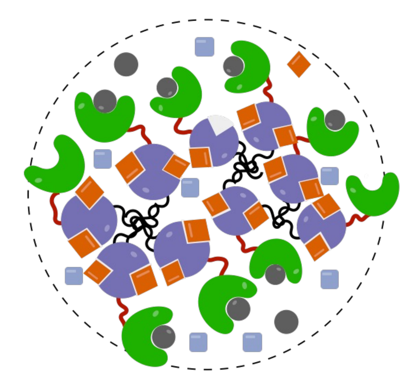


Metabolons: Intracellular metabolic assemblies



An et al., *Science* 2008

Advantages



– Increase reaction flux, pathway acceleration

– Minimize and localize unwanted intermediates

– Reduce transient timescales

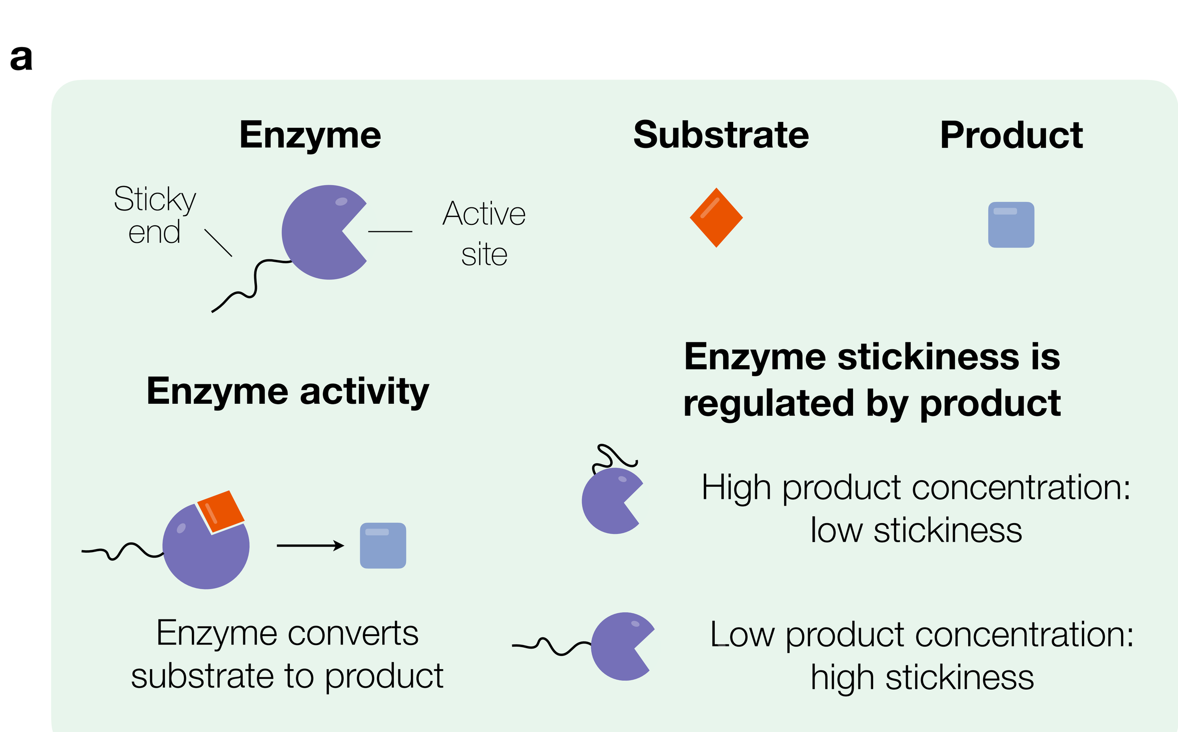
Castellana *et al.*, *Nat. Biotech.* 2014

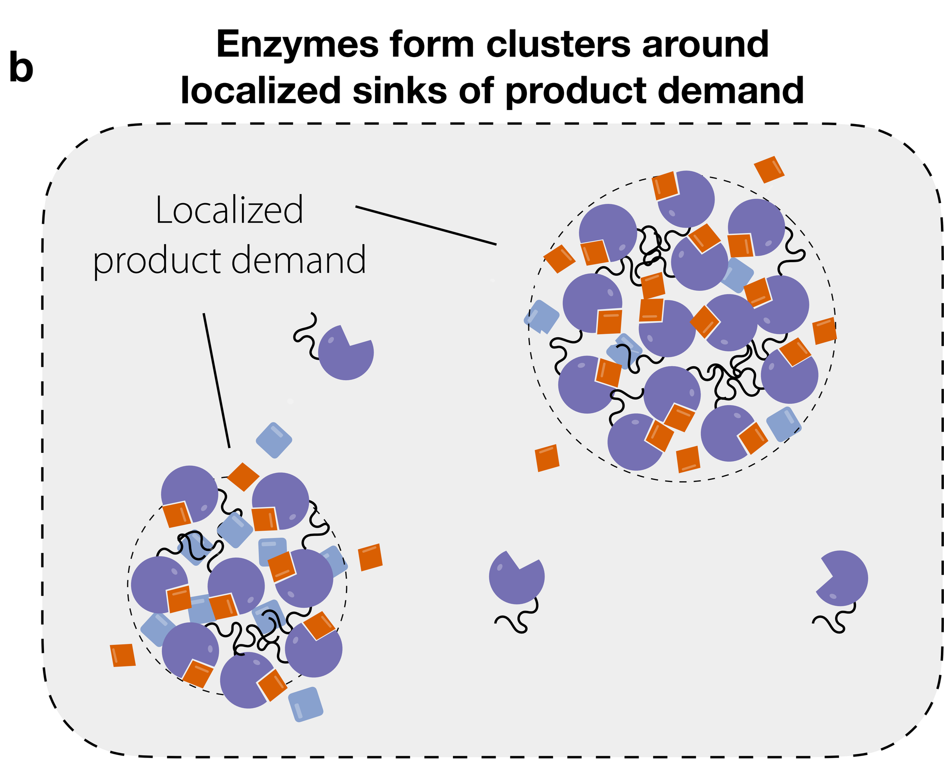
What determines the optimal size and spatial organization of condensates to maximize efficiency?

Overarching question: can enzymes self-organize into functional clusters that **maximize metabolic yield** and adapt to changes in cellular need?

Our proposal: functionally adaptive self-organization should depend on **product demand**. In brief, enzymes should only cluster when and where they are needed to make product.

Enzyme clustering via product-demand-induced phase separation





Three volume fractions: $i = \{\mathrm{s}, \mathrm{ns}, \mathrm{sol}\}$ (sticky, non-sticky, solvent)

Free energy:
$$\frac{f}{k_B T} = \underbrace{\sum_i \phi_i \ln \phi_i}_{\text{Entropy of mixing}} - \underbrace{\chi_{\text{s-s}} \phi_s^2}_{\text{Intermolecular interactions}} + \underbrace{\frac{\kappa}{2}(\nabla \phi_s)^2}_{\text{Interfacial energy}}$$

Volume conservation:
$$\partial_t \phi_i = \frac{D}{k_B T} \nabla \cdot [\phi_i \nabla(\mu_i - \mu_{\text{sol}})] - J_{i\to j} + J_{j \to i}$$

Out-of-equilibrium switching fluxes:
$$J_{\text{s}\to\text{ns}} = k_{\text{s}\to\text{ns}} \phi_s \frac{m_2}{m_2 + K} \quad \text{and} \quad J_{\text{ns}\to\text{s}} = k_{\text{ns}\to\text{s}} \phi_{\text{ns}} \frac{K}{m_2 + K}$$

Substrate:
$$\partial_t m_1 = \underbrace{D_m \nabla^2 m_1}_{\text{Diffusion}} + \underbrace{\alpha}_{\text{Production}} - \underbrace{\beta_1 m_1}_{\text{Decay}} - \underbrace{k_{\text{cat}} c_{\max} (\phi_s + \phi_{\text{ns}}) \frac{m_1}{m_1 + K}}_{\text{Catalysis}}$$

Product:
$$\partial_t m_2 = \underbrace{D_m \nabla^2 m_2}_{\text{Diffusion}} - \underbrace{\beta_2 m_2}_{\text{Decay}} - \underbrace{\sum_i j_i m_2 f_i(\boldsymbol{r})}_{\text{Localized demand}} + \underbrace{k_{\text{cat}} c_{\max} (\phi_s + \phi_{\text{ns}}) \frac{m_1}{m_1 + K}}_{\text{Catalytic production}}$$

Key features and quantities of our model

Governing dimensionless parameters

$$\tilde{\Gamma} \equiv \frac{D}{D_m} = \frac{\text{Enzyme diffusion}}{\text{Metabolite diffusion}} \qquad \tilde{\beta}_1 \equiv \frac{\beta_1}{k_{\text{cat}} c_{\max}/K} = \frac{\text{Substrate decay}}{\text{Substrate catalysis}} \qquad \tilde{\beta}_2 \equiv \frac{\beta_2}{k_{\text{cat}} c_{\max}/K} = \frac{\text{Product decay}}{\text{Catalytic production}}$$

$$\tilde{\alpha} \equiv \frac{\tilde{\alpha}}{c_{\max} k_{\text{cat}}} = \frac{\text{Substrate production}}{\text{Substrate catalysis}} \qquad \tilde{D} \equiv \frac{D/\ell_c^2}{k_{\text{ns}\to\text{s}}} = \frac{\text{Enzyme diffusion}}{\text{Enzyme switching}} \qquad \tilde{k} \equiv \frac{k_{\text{s}\to\text{ns}}}{k_{\text{ns}\to\text{s}}} = \frac{\text{s} \to \text{ns switching rate}}{\text{ns} \to \text{s switching rate}}$$

$$\tilde{j}_i \equiv \frac{j_i}{k_{\text{cat}} c_{\max}/K} = \frac{\text{Product demand}}{\text{Catalytic production}}$$

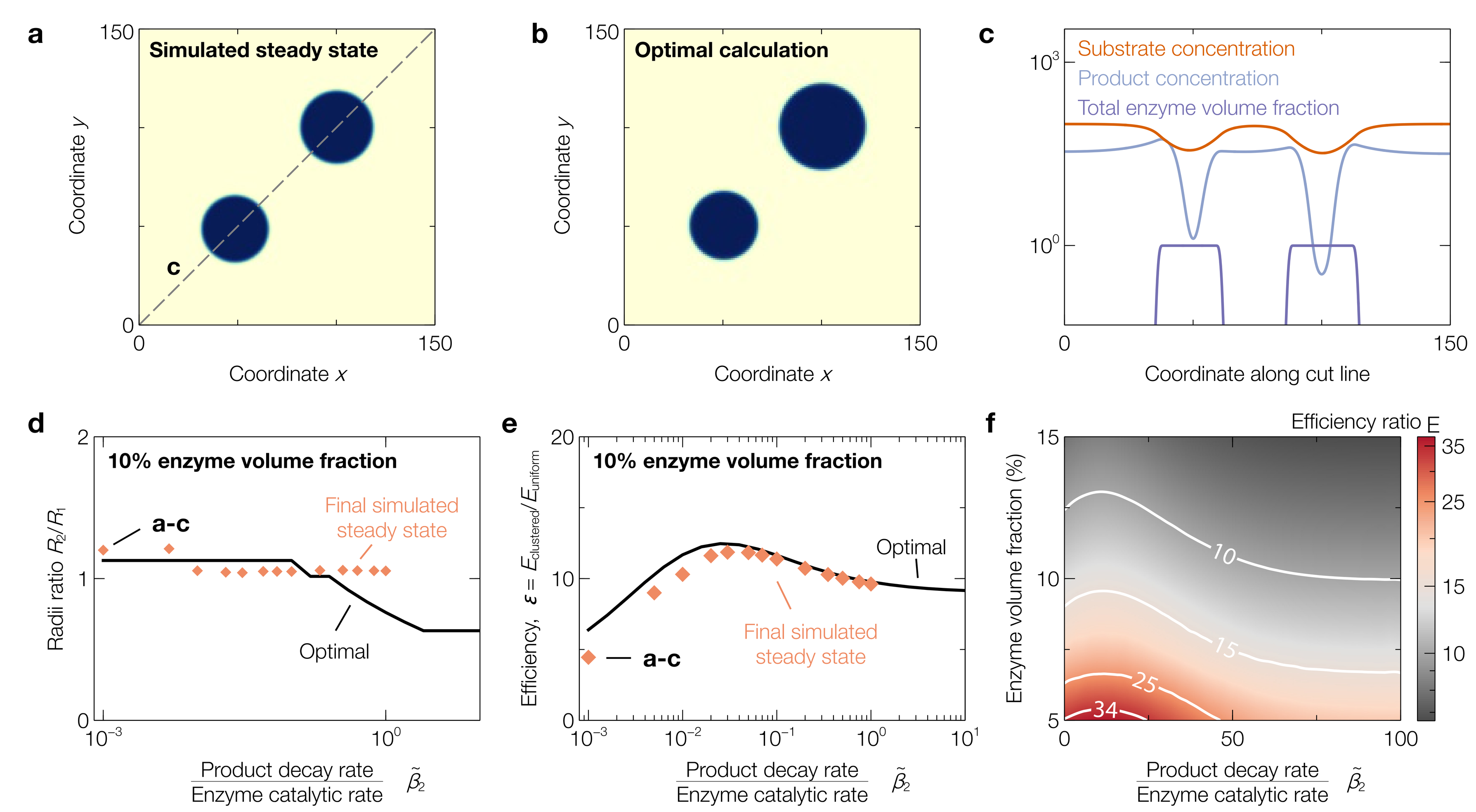
Efficiency of the pathway: $E \equiv \dfrac{\int \mathrm{d}\boldsymbol{r} j_i m_2 f_i(\boldsymbol{r})}{\int \mathrm{d}\boldsymbol{r} \beta_2 m_2} = \dfrac{\text{Product use}}{\text{Product waste}}$

Clustering advantage: $\epsilon \equiv \dfrac{E_{\text{clustered}}}{E_{\text{uniform}}}$

Biophysical constraints: $\tilde{\alpha} \sim 1, \tilde{\beta}_1 \ll 1, \tilde{D} \sim 10 - 10^2, \tilde{\Gamma} \ll 1, \tilde{k} \sim 10^{-2}, \tilde{j}_i \sim 1 - 10$

Efficiency gains are multiplicative with each step of the pathway

Product-dependent enzyme clustering around spatially localized sinks achieves nearly optimal efficiency



Main takeaways:

– Enzymes can **adapt to product demand** by self-organizing into clusters that enhance production where needed.

– Product-demand-dependent clustering can achieve **near optimal pathway efficiencies**.

– Efficiency increases by up to ~30-fold for low enzyme fractions compared to the uniform case.

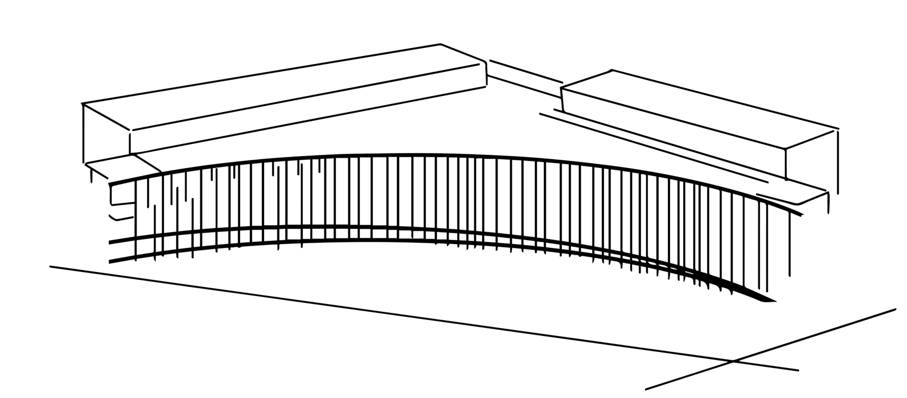
Next steps and open questions

– Explore and characterize parameter space

– Uniform product demand: arrested coarsening near optimal size?

– **Equilibrium enzyme switching:** allosteric enzymes

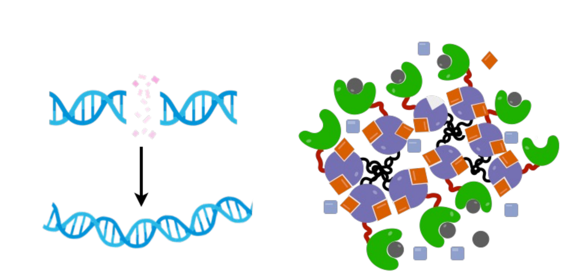
– Include reverse catalysis of product



Beyond metabolism

– Condensates at DNA repair sites

– Multicomponent interactions and complex environments



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